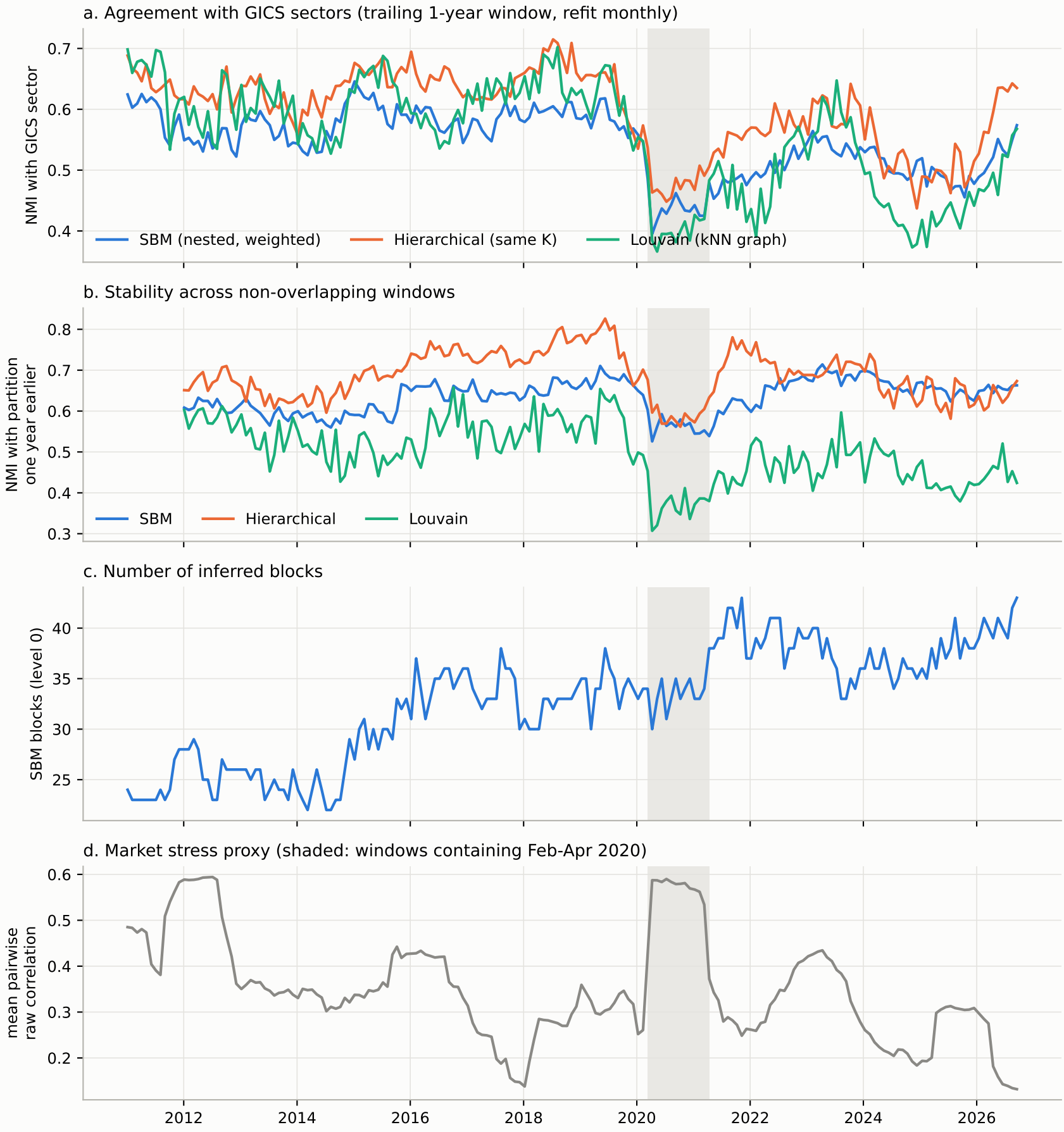
a. Agreement with GICS sectors (trailing 1-year window, refit monthly)
NMI with GICS sector
0.7
0.6
0.5
0.4
SBM (nested, weighted)
Hierarchical (same K)
Louvain (kNN graph)
b. Stability across non-overlapping windows
NMI with partition one year earlier
0.8
0.7
0.6
0.5
0.4
0.3
SBM
Hierarchical
Louvain
c. Number of inferred blocks
SBM blocks (level 0)
40
35
30
25
d. Market stress proxy (shaded: windows containing Feb-Apr 2020)
mean pairwise raw correlation
0.6
0.5
0.4
0.3
0.2
2012
2014
2016
2018
2020
2022
2024
2026